

Scaled Invariance of Infinity and Zero: The Counting Unit Is the Reference Frame

How “infinitely many” and “none” become two readings of one interval

B. Wallace (TensorRent)

with computational verification by Claude (Anthropic)

Companion to *Form/Function Relativity* (FF06g) and *When a Number Lies* — SIP License v1.1 — Seed
20260423

Abstract

There are infinitely many numbers between 1 and 2 in the decimals, and there are none between them in the whole numbers. Both statements are true, of the same pair $(1, 2)$, and the framework’s discipline is to notice that a quantity carrying two opposite values cannot be a property of the object it is attached to. It is a property of a *pair*. We make this precise: the count of representable points inside an interval is a reading of (interval, unit), where the unit δ is the minimum resolvable step — a freely chosen reference frame, exactly the role played by the surrogate ensemble in *Form/Function Relativity* (FF06g) and by the reference measure $d\mu/d\nu$ in the core monograph’s asymmetry lemma. The labels ∞ (as $\delta \rightarrow 0$) and 0 (interior of $(1, 2)$ at $\delta = 1$) are the same interval read at two units related by a scale. We isolate the invariant: the count is preserved *exactly* under the joint rescaling $([a, b], \delta) \mapsto (\lambda[a, b], \lambda\delta)$ — verified with exact rational arithmetic over 2×10^5 random configurations, 0 mismatches — and it moves under *interval-only* scaling, which is why the label is relative rather than intrinsic. The only frame-free content is the dimensionless ratio width/δ . The interior of $(1, 2)$ carries 0, 9, 99, 999, 9999 points at $\delta = 1, \frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \frac{1}{10000}$: one interval, the whole staircase from “empty” to “diverging,” driven only by the unit. This is a classical, machine-verified (T1) statement about resolution-relative counting. **It is not a claim about set cardinality:** $|\mathbb{R}|, |\mathbb{Q}|, |\mathbb{Z}|$ remain genuinely distinct and frame-independent, and Cantor is nowhere touched. The observation is the reframing; the scope boundary is the point.

Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

1 The observation

Ask how many numbers lie between 1 and 2. In the decimals the answer is “infinitely many”; in the whole numbers the answer is “none.” The two answers are not in tension — they are answers to two different questions wearing the same words. What changed between them is not the interval $[1, 2]$, which is fixed, but the *unit*: the smallest step you are permitted to resolve. The program’s first move, made already in *When a Number Lies* and *Form/Function Relativity*, is to refuse to attach a swinging label to a fixed object. If the count of interior points takes the value ∞ under one convention and 0 under another, then the count is not a property of $[1, 2]$. It is a property of the *pair* $([1, 2], \delta)$, and δ — the resolution unit — is a chosen reference frame.

2 The unit as reference frame

Definition 1 (Resolution unit and its count). Fix a unit $\delta > 0$: the minimum resolvable step, so that the only representable points are the lattice $\delta\mathbb{Z} = \{m\delta : m \in \mathbb{Z}\}$. For an interval $[a, b]$ define the closed count

$$N_\delta[a, b] = \#(\delta\mathbb{Z} \cap [a, b]) = \lfloor b/\delta \rfloor - \lceil a/\delta \rceil + 1 \quad (\text{clamped at } 0),$$

and the interior count $N_\delta^\circ(a, b) = \#(\delta\mathbb{Z} \cap (a, b))$. The “decimals” limit is $\delta \rightarrow 0$; the “whole numbers” frame is $\delta = 1$.

The count is now explicitly a two-argument object $N_\delta[a, b]$, and the content of this note is that the second argument is load-bearing and that its dependence has an exact symmetry.

Principle 1 (Scaled invariance of the counting label). *The count of representable points in an interval is a coordinate on the pair (interval $\times$ unit), not a property of the interval. It is invariant under the joint rescaling $([a, b], \delta) \mapsto (\lambda[a, b], \lambda\delta)$ for every $\lambda > 0$, and it varies under scaling either factor alone. The labels “ ∞ ” and “0” for the interior of a fixed interval are therefore relative to the unit, and are related to one another by a change of scale — infinity and zero are the two ends of one staircase in δ .*

Proposition 1 (Exact joint-scaling invariance). *For all $a \leq b$, $\delta > 0$, $\lambda > 0$,*

$$N_\delta[a, b] = N_{\lambda\delta}[\lambda a, \lambda b].$$

Proof. The map $x \mapsto \lambda x$ is a bijection $\delta\mathbb{Z} \rightarrow (\lambda\delta)\mathbb{Z}$ carrying $[a, b]$ onto $[\lambda a, \lambda b]$: $x \in \delta\mathbb{Z} \cap [a, b] \iff \lambda x \in (\lambda\delta)\mathbb{Z} \cap [\lambda a, \lambda b]$. A bijection preserves cardinality. Equivalently, both counts equal $\lfloor b/\delta \rfloor - \lceil a/\delta \rceil + 1$ because $(\lambda b)/(\lambda\delta) = b/\delta$ and $(\lambda a)/(\lambda\delta) = a/\delta$. $\square$ $\square$

Corollary 1 (The invariant is dimensionless). *$N_\delta[a, b]$ depends only on the ratios a/δ and b/δ . The raw width and the raw unit are gauge; only width/ δ is physical. In particular the interior count between two consecutive units is 0 for every δ — “adjacent” means “one unit apart,” a statement about δ , never about the two endpoints.*

Proposition 1 is the precise form of “infinity and zero are only a relative scaled invariance.” Scale the interval and the unit together and nothing moves; the count is a rigid invariant. Scale one without the other and the label slides continuously from 0 to ∞ . There is no absolute count of “the numbers between 1 and 2”; there is only the count at a stated resolution, and a symmetry that says which reparametrizations leave it fixed.

3 Measurement

The counting facts are exact integers; we verify them with exact rational arithmetic (Python `fractions`, no floating error), seed 20260423. The driver is `code/notes_verification/test_scaled_invariance.py`.

A. Unit-relativity of the label. The interior of the single fixed interval $(1, 2)$, read at shrinking units:

unit δ	1	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	$\frac{1}{10000}$
$N_\delta^\circ(1, 2)$	0	9	99	999	9999
label	empty	$\longrightarrow$ diverging ($\rightarrow \infty$)			

One interval; the entire span from “nothing between 1 and 2” to “infinitely many between 1 and 2” is produced by the unit alone. Principle 1 made numerical.

B. Joint-scaling invariance (the symmetry). Over 2×10^5 random configurations $([a, b], \delta, \lambda)$ with a, b, δ, λ rational, comparing $N_\delta[a, b]$ against $N_{\lambda\delta}[\lambda a, \lambda b]$: 0 **mismatches** (largest count encountered 45,601). Proposition 1 holds exactly, as a bijection must.

C. Interval-only scaling breaks it (why it is “relative”). Holding $\delta = 1$ and stretching the interval:

$$N_1^\circ(1, 2) = 0, \quad N_1^\circ(10, 20) = 9, \quad N_1^\circ(100, 200) = 99, \quad N_1^\circ(1000, 2000) = 999.$$

The $\times 10$ image of $[1, 2]$ gains 9 interior points *because the unit did not scale with it*. The adjacency of 1 and 2 is the statement $b - a = \delta$, a fact about the frame.

D. The invariant is dimensionless. Over 5×10^4 random ratio-preserving reparametrizations, $N_\delta[a, b] = N_{\lambda\delta}[\lambda a, \lambda b]$ in 50,000/50,000 cases, confirming Corollary 1: the count is a function of $a/\delta, b/\delta$ alone.

All four are machine-verified (T1); the script exits nonzero on any mismatch.

4 Why this is the same object as the framework’s other relativities

This is not a new mechanism; it is the framework’s standing one, applied to counting.

The unit is the reference measure. *Form/Function Relativity* (FF06g) showed the form/function verdict to be a coordinate on (witness $\times$ frame), with the frame a chosen surrogate ensemble ν — the reference measure $d\mu/d\nu$ of the core monograph’s BCH-transfer-entropy lemma. Here the “frame” is the resolution unit δ and the “label” is the cardinality reading $\{0, \dots, \infty\}$ rather than $\{\text{form, function}\}$. In both cases a scalar that looks intrinsic is exhibited as a two-argument object, and the second argument — the frame — is free. Choosing it differently relabels; that choice is the perspective.

The scalar was a shadow. *When a Number Lies* isolates the pattern that a single scalar is often a lossy projection of an underlying relation, and that the relation, not the scalar, is the faithful object. “The number of points between 1 and 2” is such a shadow: the faithful object is the relation width/ δ between the interval and the unit, and only that dimensionless ratio survives a change of frame (Corollary 1). “ ∞ ” and “0” are what the shadow reads when the unit is sent to 0 or set equal to the width.

Form/function is frame-relative; so is cardinality-under-resolution. The parent program’s thesis is that form and function are generated by the asymmetry of a pair and carried by neither member alone. Principle 1 is the counting instance: “how many” is generated by the asymmetry between an interval and a unit, and is carried by neither the interval nor the unit alone.

5 What this establishes, and where it stops

Establishes. For a fixed interval, the count of representable points is frame-relative in the exact sense of Definition 1: it ranges over $0, 9, 99, 999, 9999, \dots$ on the single interval $(1, 2)$ as the unit shrinks (measurement A), it is preserved *exactly* under joint rescaling of interval and unit (Proposition 1, $0/2 \times 10^5$ mismatches), it changes under interval-only scaling (measurement C), and its only frame-free content is the dimensionless ratio width/δ (Corollary 1, $50000/50000$). “Infinity” and “zero” are the $\delta \rightarrow 0$ and $\delta = \text{width}$ ends of one staircase, related by a scale.

Does not establish — the load-bearing boundary. This is a statement about the count of points *representable at a resolution*, i.e. the lattice $\delta\mathbb{Z}$. It is **not** a statement about set cardinality. The claim “there are infinitely many reals between 1 and 2” is true and frame-*independent* as a statement about $\mathbb{R}$: $|[1, 2] \cap \mathbb{R}| = \mathfrak{c}$ and $|[1, 2] \cap \mathbb{Q}| = \aleph_0$ regardless of any unit, because those sets contain their points with no resolution imposed. Cantor’s theorem, the distinctness of $\aleph_0$ and $\mathfrak{c}$, and the density of $\mathbb{Q}$ in $\mathbb{R}$ are untouched and uncontested here. The relativity is precisely the difference between “*how many points exist in the set*” (cardinality; frame-free; the reals are genuinely uncountable) and “*how many points are resolvable under a minimum step δ* ” (a count; frame-relative; what this note is about). Conflating the two would be the overclaim, and the boundary between them *is* the result: the whole-number constraint does not make the reals disappear — it selects a coarser frame in which the interior of $(1, 2)$ is empty, and Proposition 1 says exactly how that frame relates to the finer ones.

Two honest edges. (i) $N_\delta^\circ(1, 2) = k - 1$ at $\delta = 1/k$ grows without bound but is finite at every $\delta > 0$; “ ∞ ” is the limit $\delta \rightarrow 0$, not a value at any admissible unit, and the limit is exactly where the resolution frame is abandoned and the set-cardinality question (a different question) takes over. (ii) The invariance is a discrete lattice fact, not a measure-theoretic one; the continuous analogue — Lebesgue length $b - a$ scaling by λ under $x \mapsto \lambda x$ — is the $\delta \rightarrow 0$ shadow of Proposition 1 with the joint rescaling of the unit made explicit rather than hidden. Both edges are recorded so the staircase is not over-read.

This companion accompanies *Form/Function Relativity* (FF06g) and *When a Number Lies*. All counts are exact-integer and reproduced standalone via `code/notes_verification/test_scaled_invariance.py` at seed 20260423. SIP License v1.1.

Amendment: scope and evidence, 24 September 2026

A finite-resolution count and its joint-scaling ratio are properties of a specified observation protocol. They do not replace mathematical zero, infinity or quantification over an infinite domain. In the action study, dimensional scaling preserves mass ratios while changing the physical scale; this is a family of actions, not proof that a dimensionful observable is invariant. Any limit claim needs its own topology, norm and convergence assumptions.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.